# Bilastine and Montelukast Tablets

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Manufacturers, regulatory authorities, health authorities, researchers, and other stakeholders are invited to provide their feedback and comments on this draft proposal. Manufacturers are also invited to submit samples of their products to the IPC to ensure that the proposed monograph adequately controls the quality of the product(s) they manufacture. Comments and samples received after the last date will not be considered by the IPC before finalizing the monograph.

**Please send any comments you may have on this draft document to lab.ipc@gov.in, with a copy to Dr. Gaurav Pratap Singh (email: gpsingh.ipc@gov.in) before the last date for comments.**

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## Document History and Schedule for the Adoption Process

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| Further follow-up action as required. | |

# Bilastine and Montelukast Tablets

Bilastine and Montelukast Sodium Tablets

Bilastine and Montelukast Tablets contain montelukast sodium equivalent to not less than 90.0 per cent and not more than 110.0 per cent of the stated amount of montelukast, $C_{35}H_{36}ClNO_3S$ and bilastine, $C_{28}H_{37}N_3O_3$.

**Usual strength**. Bilastine, 20 mg and Montelukast, 10 mg.

## Identification

In the Assay, the principal peaks in the chromatogram obtained with the test solution correspond to the principal peaks in the chromatogram obtained with the reference solution.

## Tests

**Dissolution** (2.5.2).

*For Bilastine —*

Apparatus No. 2 (Paddle),
Medium. 900 ml of *0.1 M hydrochloric acid*,
Speed and time. 50 rpm and 45 minutes.

Withdraw a suitable volume of the medium and filter.

*Test solution.* Use the filtrate, dilute if necessary, with the dissolution medium.

*Reference solution (a)*. A 0.022 per cent w/v solution of *bilastine IPRS* in *methanol*. Dilute 5.0 ml of the solution to 50.0 ml with the dissolution medium.

Chromatographic system
- a stainless steel column 25 cm x 4.6 mm, packed with octadecylsilane bonded to porous silica (5 µm) (Such as Hypersil BDS),
- column temperature: 40°,
- mobile phase A. a 0.1 per cent v/v solution of *triethylamine* in *water*, adjusted to pH 3.0 with *orthophosphoric acid*, B. *acetonitrile*,
- a gradient programme using the conditions given below,
- flow rate: 1 ml per minute,
- spectrophotometer set at 220 nm,
- injection volume: 10 µl.

| Time (in min.) | Mobile phase A (per cent v/v) | Mobile phase B (per cent v/v) |
|---|---|---|
| 0 | 45 | 55 |
| 2 | 45 | 55 |
| 18 | 10 | 90 |
| 20 | 45 | 55 |
| 25 | 45 | 55 |

Inject the reference solution. The test is not valid unless the column efficiency is not less than 2000 theoretical plates, the tailing factor is not more than 2.0 and the relative standard deviation for replicate injections is not more than 2.0 per cent.

Inject the reference solution and the test solution.

Calculate the content of $C_{28}H_{37}N_3O_3$ in the medium.

Q. Not less than 70 per cent of the stated amounts of $C_{28}H_{37}N_3O_3$.

*For Montelukast —*

*NOTE —Protected the solutions from light.*

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Apparatus No. 2 (Paddle),
Medium. 900 ml of 0.5 per cent w/v *sodium lauryl sulphate* in *water*,
Speed and time. 50 rpm and 45 minutes.

Withdraw a suitable volume of the medium and filter.

*Test solution*. Use the filtrate, dilute if necessary, with the dissolution medium.

*Reference solution.* A solution of *montelukast sodium IPRS* containing 0.022 per cent w/v of montelukast in *methanol*. Dilute 5.0 ml of the solution to 100.0 ml with the dissolution medium.

Use chromatographic ystem as described under Dissolution for Bilastine.

Inject the reference solution. The test is not valid unless the column efficiency is not less than 2000 theoretical plates, the tailing factor is not more than 2.0 and the relative standard deviation for replicate injections is not more than 2.0 per cent.

Inject the reference solution and the test solution.

Calculate the content of $C_{35}H_{36}ClNO_3S$ in the medium.

Q. Not less than 70 per cent of the stated amounts of $C_{35}H_{36}ClNO_3S$.

**Related substances**. Determine by liquid chromatography (2.4.14).

*NOTE — Prepare solutions immediately before use and protected from light.*

*Solvent mixture.* 0.1 per cent v/v solution of *triethylamine* in *methanol*.

*Test solution.* Disperse a quantity of powdered tablets containing 100 mg of montelukast in 20 ml of *water* with the aid of ultrasound for 5 minutes, add 100 ml of the solvent mixture and further sonicate for 20 minutes with intermittent shaking, dilute to 200.0 ml with the solvent mixture, filter. *(NOTE- Maintain the sonicator bath temperature 10°)*.

*Reference solution*. A solution of *montelukast sodium IPRS* containing 0.025 per cent w/v of montelukast and 0.05 per cent w/v of *bilastine IPRS* in the solvent mixture. Dilute 1.0 ml of the solution to 100.0 ml with the solvent mixture.

Chromatographic system
- a stainless steel column 15 cm x 4.6 mm, packed with octadecylsilane bonded to porous silica (5 µm) (Such as Hypersil ODS),
- column temperature: 40°,
- mobile phase: A. a 0.6 per cent w/v solution of *ammonium acetate* in *water*, adjusted to pH 5.5 with *acetic acid*,
  B. *methanol*,
- a gradient programme using the conditions given below,
- flow rate: 1 ml per minute,
- spectrophotometer set at 240 nm,
- injection volume: 10 µl.

| Time (in min.) | Mobile phase A (per cent v/v) | Mobile phase B (per cent v/v) |
|---|---|---|
| 0 | 50 | 50 |
| 15 | 35 | 65 |
| 40 | 30 | 70 |
| 60 | 20 | 80 |
| 65 | 20 | 80 |
| 70 | 50 | 50 |
| 75 | 50 | 50 |

| Name | Relative retention time |
|---|---|
| Montelukast sulphoxide impurity | 0.6 |
| Montelukast | 1.0 |
| Montelukast styrene impurity | 1.4 |

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Inject the reference solution. The test is not valid unless the column efficiency is not less than 2000 theoretical plates, tailing factor is not more than 2.0 and the relative standard deviation for replicate injections is not more than 5.0 per cent, for both the peaks.

Inject the reference solution and the test solution. In the chromatogram obtained with the test solution, the area of any peak corresponding to montelukast sulphoxide is not more than 3 times the area of the montelukast peak in the chromatogram obtained with the reference solution (1.5 per cent), the area of any peak corresponding to montelukast styrene is not more than the area of the montelukast peak in the chromatogram obtained with the reference solution (0.5 per cent), the area of any other secondary peak is not more than the area of the montelukast peak in the chromatogram obtained the reference solution (0.5 per cent) and the sum of areas of all the secondary peaks is not more than 6 times the area of the montelukast peak in the chromatogram obtained with the reference solution (3.0 per cent).

**Uniformity of dosage units** (2.5.4). Comply the tests stated under Tablets.

**Other tests**. Comply with the tests stated under Tablets.

**Assay**. Determine by liquid chromatography (2.4.14)

*NOTE —Protected the solutions from light.*

*Solvent mixture.* Equal volumes of *acetonitrile* and *water*.

*Test solution.* Weigh and powder 20 tablets. Disperse a quantity of the powder containing 100 mg of montelukast in 170 ml of the solvent mixture with the aid of ultrasound for 20 minutes with intermittent shaking and dilute to 250.0 ml with the solvent mixture. Dilute 5.0 ml of the solution to 50.0 ml with the solvent mixture. *(NOTE- Maintain the sonicator bath temperature 10°).*

*Reference solution*. A solution of *montelukast sodium IPRS* containing 0.04 per cent w/v of montelukast and 0.08 per cent w/v of *bilastine IPRS* in the solvent mixture. Dilute 5.0 ml of the solution to 50.0 ml with the solvent mixture.

Use chromatographic system as described under Dissolution for Bilastine.

Inject the reference solution. The test is not valid unless the column efficiency is not less than 2000 theoretical plates, the tailing factor is not more than 2.0 and the relative standard deviation for replicate injections is not more than 2.0 per cent, for both the peaks.

Inject the reference solution and the test solution.

Calculate the content of $C_{35}H_{36}ClNO_3S$ and $C_{28}H_{37}N_3O_3$ in the tablets.

**Storage**. Store protected from light and moisture, at a temperature not exceeding 30°.

**Labelling**. The label states the strength in terms of the equivalent amount of Montelukast and Bilastine.

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